P R O J E C T R E P O R T

Customer Segmentation & Churn Pattern Analytics

in European Banking

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10,000 Customer Records | France · Germany · Spain | Python · Pandas · Seaborn

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1. Problem Statement

Customer churn is a silent revenue leak. In European retail banking, every departure erases future income, increases acquisition costs, and shrinks customer lifetime value. At a 20% churn rate across 10,000 customers, that's 2,037 lost relationships per cycle — yet most banks still respond with generic, reactive strategies that miss the real problem entirely.

> **The core challenge: banks have rich data but treat churn as a single number. This project reframes it as a segmented phenomenon — revealing exactly who is leaving, why, and where intervention will make the greatest impact.**

2. Introduction

This report presents a full Exploratory Data Analysis (EDA) of customer churn across a European bank operating in France, Germany, and Spain, using a 10,000-record dataset with 14 variables. The analysis is designed for both technical and business audiences, with one unified goal: move beyond the aggregate churn rate and understand attrition as a structured, segment-level problem that can be targeted and prevented.

The overall churn rate of 20.37% is the starting point — not the conclusion. As the analysis reveals, that single figure conceals extreme variation: a 32% churn rate in Germany, a 51% rate among customers aged 46–60, and a 100% rate among customers holding four products. These are not anomalies. They are patterns — and patterns can be acted upon.

3. Objectives

This analysis aims to: (1) decompose the overall churn rate across geographic, demographic, behavioral, and financial segments; (2) identify the highest-risk customer clusters and any multi-variable interaction effects; (3) challenge common retention assumptions — product bundling, tenure, credit quality — against actual data; and (4) translate every finding into a concrete, actionable business recommendation.

4. Dataset Description

The dataset contains 10,000 customer records from 2025 with 14 variables — zero missing values, confirmed binary column integrity, and a churn rate of 20.37% (2,037 churned, 7,963 retained). Key statistics: median age 37, median balance €97,199, median salary €100,194. Geography split: France 50.1%, Germany 25.1%, Spain 24.8%.

Variable	Type	Description
CreditScore	Integer	Credit score (350–850)
Geography	Categorical	France, Germany, or Spain
Gender	Categorical	Male / Female
Age	Integer	Customer age (18–92)
Tenure	Integer	Years with bank (0–10)
Balance	Float	Account balance in euros

Variable	Type	Description
NumOfProducts	Integer	Banking products held (1–4)
HasCrCard	Binary	1 = holds credit card
IsActiveMember	Binary	1 = active member
EstimatedSalary	Float	Annual salary estimate
Exited	Binary	Target: 1 = churned

5. Exploratory Data Analysis (EDA)

5.1 Data Ingestion & Validation

The dataset was loaded and validated for completeness, correct data types, and binary column integrity. All 14 columns returned zero nulls. The active/inactive split was near-even: 5,151 active (51.5%) vs. 4,849 inactive (48.5%), making activity status a potentially strong churn signal.

```
df = pd.read_csv('European_Bank.csv')
df.shape          # (10000, 14)
df.isnull().sum() # All zeros
df['Exited'].value_counts(normalize=True).mul(100).round(2)
# 0: 79.63%   1: 20.37%
```

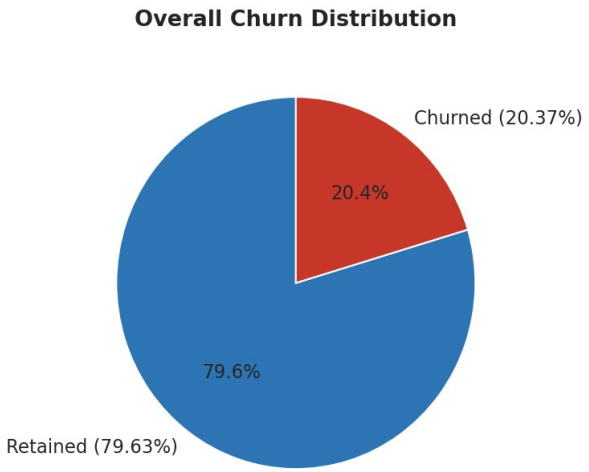

Figure 1: Overall churn distribution — 1 in 5 customers has exited

5.2 Data Cleaning & Feature Engineering

CustomerId and Surname were dropped. Four continuous variables were binned into business-relevant segments to unlock behavioral patterns invisible in raw numbers.

```
df = df.drop(['CustomerId','Surname'], axis=1)
df['AgeGroup']      = pd.cut(df['Age'], [18,30,45,60,100], labels=['<30','30-45','46-
60','60+'])
df['CreditBand']    = pd.cut(df['CreditScore'], [300,580,700,850],
labels=['Low','Med','High'])
df['TenureGroup']   = pd.cut(df['Tenure'], [-1,3,7,10], labels=['New','Mid','Long'])
df['HighValue']     = df['Balance'] > 100_000
```

5.3 Geographic Segmentation

Geography emerged as the most immediately actionable variable. Germany's churn rate of 32.44% is nearly double France (16.15%) and Spain (16.67%) — a 12-percentage-point gap that signals a systemic, country-level problem rather than random variation.

```
(df.groupby('Geography')['Exited'].mean()*100).round(2)
# France: 16.15   Germany: 32.44   Spain: 16.67
```

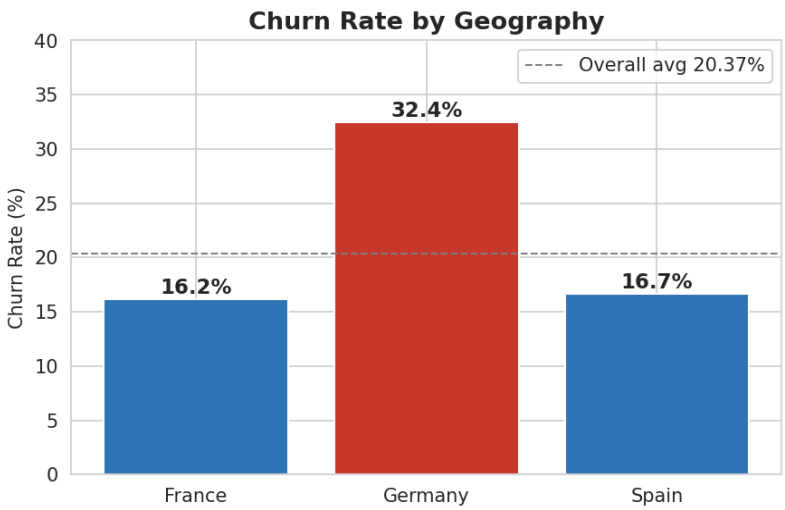

Figure 2: Churn rate by geography — Germany at 32.44% is the clear outlier

Geography	Churn Rate	vs. Overall Avg
France	16.15%	−4.22 pp
Germany	32.44%	+12.07 pp
Spain	16.67%	−3.70 pp

5.4 Age Group Segmentation

Age is the strongest single demographic predictor in the dataset. The 46–60 cohort churns at 51.12% — the majority of that age group has already left. These are peak-earning customers with complex financial needs; their departure signals a critical advisory and service gap.

```
(df.groupby('AgeGroup')['Exited'].mean()*100).round(2)
# <30: 7.50   30-45: 15.74   46-60: 51.12   60+: 24.78
```

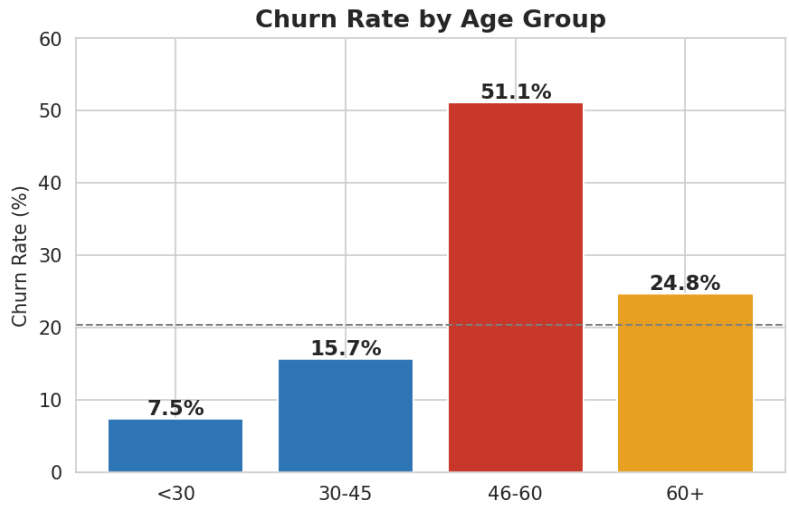

Figure 3: Churn by age group — 46–60 cohort exceeds 51%

5.5 Product Usage vs. Churn

This is the most counterintuitive finding of the analysis. Two products is the loyalty sweet spot (7.58% churn). Beyond that, churn escalates catastrophically — 82.71% for three products, and 100% for four. The conventional cross-selling retention hypothesis fails entirely here.

```
(df.groupby('NumOfProducts')['Exited'].mean()*100).round(2)
# 1: 27.71   2: 7.58   3: 82.71   4: 100.00
```

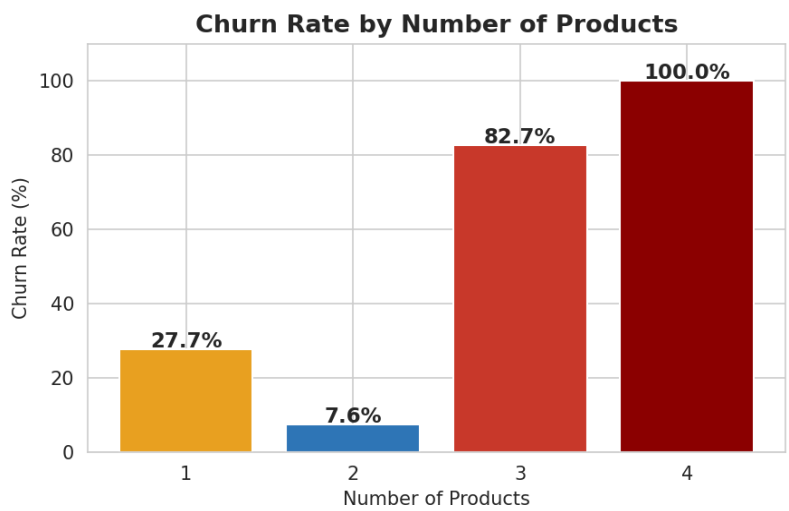

Figure 4: Churn by product count — dramatic non-linear escalation beyond 2 products

Products	Churn Rate	Signal
1	27.71%	Above average — single-touch relationship
2	7.58%	Loyalty optimum — sweet spot
3	82.71%	Critical — severe friction

Products	Churn Rate	Signal
4	100.00%	Total attrition — every customer churned

Every customer holding 4 products has churned. The bank's most heavily cross-sold customers are its most dissatisfied.

5.6 Gender & Activity Status

Female customers churn at 25.07% vs. 16.46% for males — an 8.61 pp gap that persists across geographies and is unlikely to reflect financial differences. This points to a differential service experience that qualitative research must investigate.

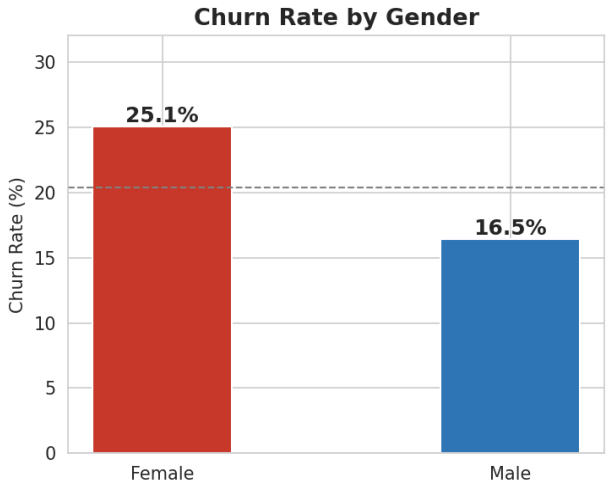

Figure 5: Female customers churn 8.6 pp higher than male customers

Inactive members churn at 26.85% vs. 14.27% for active members. The active-to-inactive transition is a leading behavioral indicator of impending churn and an ideal trigger for proactive outreach.

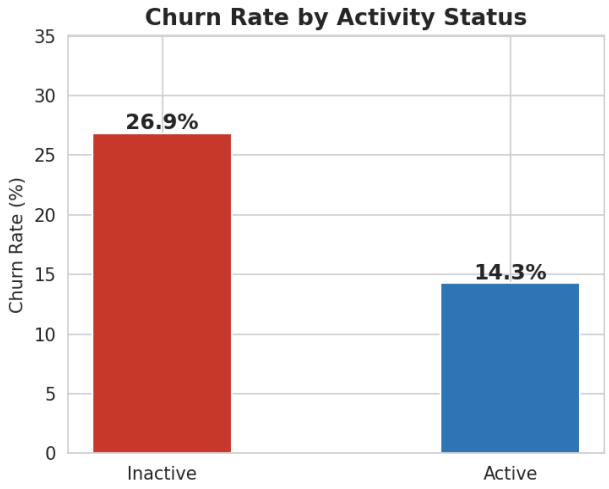

Figure 6: Inactive members churn at nearly double the rate of active members

5.7 Geography × Age Interaction

When geography and age are crossed, the highest-risk cell becomes unmistakable: German customers aged 46–60 churn at 67.3% — two-thirds of that segment has already exited. The co-occurrence of these two risk factors produces a compounded attrition effect far exceeding either variable alone.

```
ct = pd.crosstab(df['AgeGroup'], df['Geography'],
```

```
                 values=df['Exited'], aggfunc='mean')
sns.heatmap(ct, annot=True, cmap='RdYlBu_r')
```

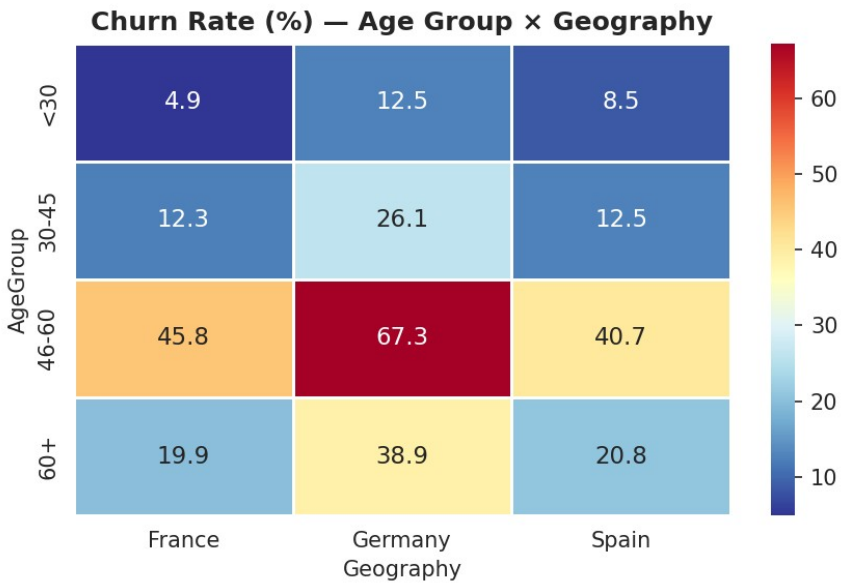

Figure 7: Churn heatmap — Germany × 46–60 at 67.3% is the single highest-risk segment

Age Group	France	Germany	Spain
< 30	4.9%	12.5%	8.5%
30–45	12.3%	26.1%	12.5%
46–60	45.8%	67.3%	40.7%
60+	19.9%	38.9%	20.8%

Germany × 46–60 at 67.3% is the single most critical segment in this dataset. This is a complete retention failure for that group.

5.8 Salary & High-Value Customer Analysis

Salary shows zero predictive power — churned and retained customers have near-identical salary distributions (median ~€100,193 vs. €100,194). Churn is not a financial distress problem; it is an experience and service quality problem.

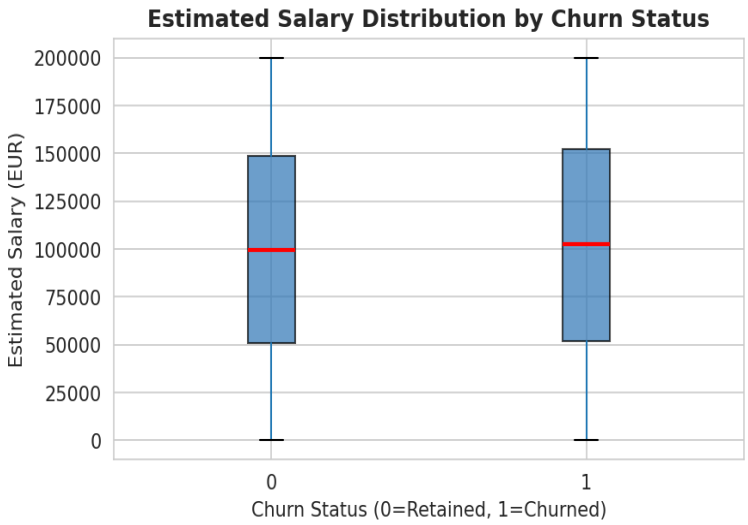

Figure 8: Salary distributions for churned vs. retained customers are virtually identical

High-value customers (balance > €100,000) churn at 25.23% — 9.35 pp above standard customers (15.88%). The bank's highest-balance customers are its most at-risk, making high-value retention one of the highest-ROI interventions available.

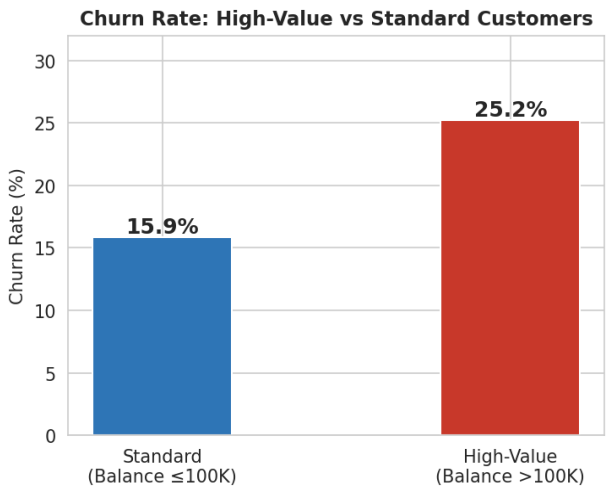

Figure 9: High-value customers churn at 25.23% vs. 15.88% for standard customers

Segment	Definition	Churn Rate	Gap
Standard	Balance ≤ €100K	15.88%	—
High-Value	Balance > €100K	25.23%	+9.35 pp

6. Key Insights

Six findings emerge with consistent clarity across all segments of the analysis:

1. Germany is the churn epicenter.

At 32.44% — double France and Spain — Germany demands a dedicated, country-level retention response, not a generic fix.

2. The 46–60 cohort is in crisis.

A 51.12% churn rate means the majority of mid-life customers are gone. In Germany specifically, that number reaches 67.3%.

3. Cross-selling beyond 2 products destroys retention.

The 7.58% churn rate at two products collapses to 82.71% at three and 100% at four. The multi-product retention hypothesis is empirically false for this institution.

4. Female customers are being underserved.

An 8.61 pp churn gap between female and male customers cannot be explained by financial factors. It signals an experiential gap requiring targeted qualitative investigation.

5. Inactivity predicts departure.

The active-to-inactive transition nearly doubles churn risk (14.27% → 26.85%) and is an actionable early-warning trigger for proactive retention outreach.

6. High-value customers are the highest-risk.

At 25.23% churn, the bank's most financially valuable customers are leaving at above-average rates. Salary is irrelevant — this is a service expectation problem, not an affordability one.

7. Recommendations

1 — Launch a Germany Retention Task Force

Establish a cross-functional team focused exclusively on Germany. Conduct exit interviews with recently churned German customers, benchmark fees and service standards against Sparkassen and local fintech competitors, and audit the in-branch and digital experience for the 46–60 cohort. Deliverable: a Germany-specific action plan within 90 days.

2 — Redesign the Cross-Selling Strategy

Immediately introduce a product suitability gate before offering any third product. Review existing three and four-product holders for unwanted or unused products. The strategic target: maximize customers in the two-product sweet spot. Stop selling products; start solving needs.

3 — Build a High-Value Retention Program

Assign dedicated relationship managers to all customers with balances above €100,000. Mandate quarterly proactive reviews, offer exclusive fee structures, and provide human-led financial planning — not digital-only service. A 5 pp churn reduction in this segment alone preserves millions in annual balance sheet value.

4 — Deploy an Early-Warning Detection System

Operationalize the risk factors identified here — Germany, age 46–60, 3+ products, balance >€100K, inactive status — as a rule-based alert system. Any customer matching two or more triggers should prompt relationship manager contact within five business days. Monitor the active-to-inactive transition in real time as the primary behavioral signal.

5 — Investigate the Gender Experience Gap

Commission qualitative research — focus groups and targeted satisfaction surveys — specifically for female customers, with emphasis on the 46–60 cohort. Probe communication quality, advisory relevance, and product suitability. Feed findings directly into service protocols and communication standards.

> **Churn in this institution is not random — it is concentrated, predictable, and preventable. The segments are identified. The signals are clear. The next move belongs to the bank.**

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Dataset: European Bank Customer Data (10,000 records) | Tools: Python · Pandas · NumPy · Seaborn · Matplotlib